

InfStones Email Marketing System PRD

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InfStones Email Marketing System — 基于 Plunk 开源项目二次开发，替换 MailerLite。本文档涵盖技术架构设计、开发路线图、Sprint 排期与里程碑规划。

Last updated: 2026-04-14

Deadline: 2026-07-10 (MailerLite 订阅到期)

1 项目概述

a) 项目背景：当前使用 MailerLite 进行 Newsletter 和营销邮件发送（~1K 订阅者，峰值 1W+），存在成本和数据控制问题。MailerLite 订阅已取消，2026-07-10 前必须完成替换。

b) 解决方案：基于开源项目 [Plunk](#) 进行二次开发，自托管部署，邮件发送对接现有 AWS SES。

c) 核心目标：

- 完全替代 MailerLite 的 Newsletter 和营销邮件能力
- 数据完全自有，不依赖第三方 SaaS
- 技术栈与团队一致 (React + Node.js + TypeScript)
- 支持 1W+ 订阅者规模

2 技术架构设计

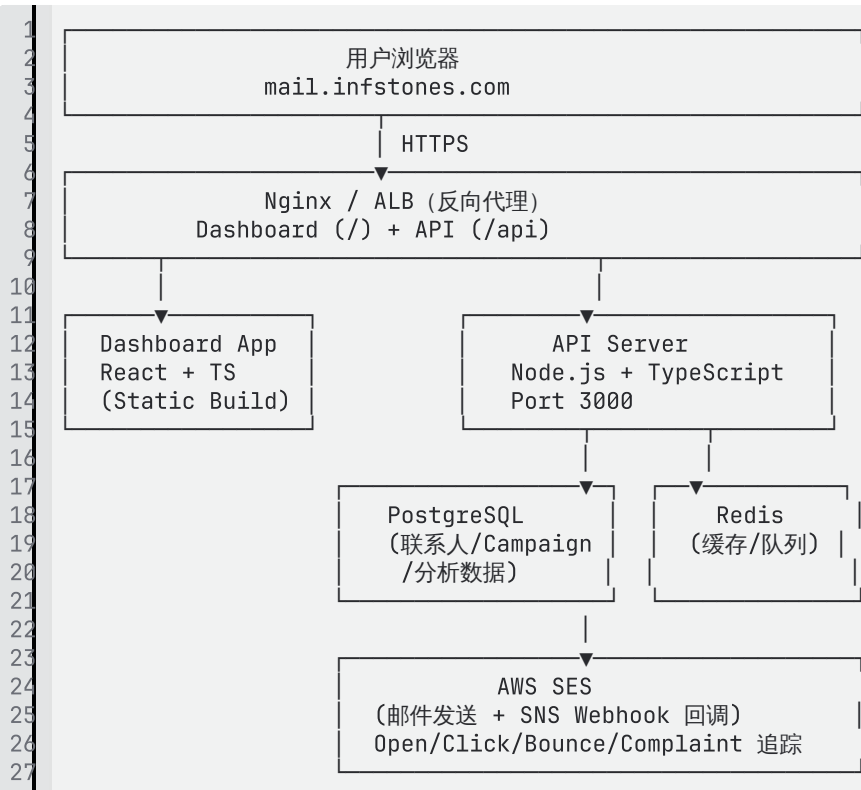
2.1 基座项目：Plunk

维度	详情
GitHub	useplunk/plunk (5K+ Stars , AGPL-3.0)
语言	TypeScript (96.1%)
后端	Node.js + TypeScript
前端	React + TypeScript
数据库	PostgreSQL
缓存	Redis
邮件发送	AWS SES (原生集成)
构建工具	Turborepo (monorepo 编排)
包管理	Yarn
测试框架	Vitest

2.2 项目结构 (Monorepo)

```
1 plunk/
2   |
3   |--- apps/
4   |   |--- api/           # 后端 API 服务 (Node.js + TypeScript)
5   |   |--- dashboard/    # 前端管理面板 (React + TypeScript)
6   |   |--- wiki/         # 文档站点 (MDX)
7   |--- packages/         # 共享库 (工具函数、类型定义等)
8   |--- docker/           # Docker 配置文件
9   |--- test/             # 测试目录
10  |--- turbo.json         # Turborepo 配置
11  |--- .yarnrc.yml        # Yarn 配置
12  |--- vitest.config.ts   # Vitest 测试配置
```

2.3 部署架构



2.4 AWS SES 集成方案

- a) 发送链路：Plunk API → AWS SES SDK → 收件人。Plunk 原生构建在 SES 之上，零迁移成本。
- b) 事件追踪：SES → SNS Topic → Plunk Webhook Endpoint。追踪 Delivery / Bounce / Complaint / Open / Click 事件。
- c) 域名验证：配置 DKIM / SPF / DMARC 记录，使用公司域名发送。
- d) 环境变量：

```
1 AWS_ACCESS_KEY_ID=<SES Access Key>
2 AWS_SECRET_ACCESS_KEY=<SES Secret Key>
3 AWS_REGION=<SES Region>
4 AWS_SES_CONFIGURATION_SET=<Configuration Set Name>
5 DATABASE_URL=postgresql://<user>:<pass>@<host>:5432/plunk
6 REDIS_URL=redis://<host>:6379
```

```
7 JWT_SECRET=<random-secret>
8 APP_URI=https://mail.infstones.com
```

3 功能清单（全部 P0）

3.1 Plunk 已具备能力（开箱即用）

功能	Plunk 现状	对应 MailerLite 步骤
Campaign (Broadcast) 创建与发送	支持一次性群发	Step 2
拖拽/可视化邮件编辑器	内置拖拽编辑器 + HTML 导入	Step 3
个性化变量替换	支持模板变量	Step 4
Campaign 详情设置	主题行、发件人信息	Step 5
订阅者管理 + 分段	基于 metadata 和行为的 Segmentation	Step 6
定时发送	Broadcast 支持 Schedule	Step 8
发送分析 (Open/Click/Bounce)	实时追踪	Dashboard
自定义域名 (DKIM/SPF)	内置支持	—

3.2 需要二开的功能（全部 P0）

功能	优先级	工作量估算	说明
Branding Customization	P0	2 天	Logo、配色、域名替换、邮件 Footer 品牌定制（全英文界面，无需汉化）

邮件预览 (Desktop + Mobile)	P0	5 天	新建响应式预览组件，支持桌面端和移动端视图切换
测试邮件发送	P0	3 天	编辑器内一键发送测试邮件到指定地址
UTM 参数管理面板	P0	5 天	Campaign 详情页新增 UTM 配置 UI，自动追加到所有链接
带 UTM 的邮件预览	P0	3 天	预览中实时显示带 UTM 参数的完整链接
邮件模板库	P0	5 天	10+ 预设模板 (Newsletter、产品公告、促销等)
编辑器体验增强	P0	5 天	增强 Block 类型 (Social Links、Button 样式、Divider)，改善拖拽流畅度
Dashboard 性能概览优化	P0	3 天	Campaign 对比、趋势图、KPI 卡片
用户管理	P0	3 天	Settings 内 Users 页面，支持添加/删除用户，Admin/Member 角色区分，顶栏当前用户标识

二开总工作量：约 33 人天。

4 开发路线图

核心目标：4 个 Sprint 内完成 MailerLite 全面替换，所有功能均为 P0。

里程碑	Sprint	日期	交付物	状态
M1: 全部功能开发完成	Sprint 1	04-14 → 04-27	环境搭建 + 所有二开功能开发完成，本地全流程可用	IN PROGRESS
M2: 部署上线测试	Sprint 2	04-28 → 05-11	Staging + Production 环境部署，QA 全量测试，UAT 验收	NOT STARTED
M3: 数据迁移	Sprint 3	05-12 → 05-25	MailerLite 全量数据迁移到 Production，数据校验通过	NOT STARTED
M4: 正式切换	Sprint 4	05-26 → 06-08	DNS 切换，团队培训，MailerLite 正式下线，进入稳定运行	NOT STARTED

06-09 ~ 07-10 为充裕的缓冲期（约 30 天），应对切换后的问题修复和体验优化。

5 Sprint 排期详细计划

5.1 Sprint 1：全部功能开发（04-14 → 04-27）

Sprint Goal：环境搭建完成，所有 P0 二开功能开发完毕，本地端到端流程全部跑通。

任务	估算	依赖
----	----	----

Fork Plunk 仓库，搭建本地开发环境 (Docker Compose)	1d	无
配置 AWS SES (IAM、SNS Topic、Configuration Set)	1d	无
端到端发送验证 (API → SES → 收件 → Webhook 回调)	1d	SES 配置
Branding Customization (Logo、配色、域名、邮件 Footer)	2d	本地环境
邮件预览组件开发 (Desktop + Mobile 视图切换)	3d	本地环境
测试邮件发送功能 (编辑器内一键发送到指定地址)	2d	预览组件
UTM 参数管理 UI + 自动追加逻辑	3d	无
带 UTM 的邮件预览	1d	UTM 逻辑 + 预览组件
编辑器 Block 增强 (Social Links、Button 样式、Divider)	3d	无
邮件模板库 (10+ 预设模板)	3d	编辑器增强
Dashboard 性能概览优化 (KPI 卡片、趋势图、Campaign 对比)	2d	无
CI/CD Pipeline 搭建 (Build → Test →	1d	无

Deploy)		
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说明：多项任务可并行开发。建议至少 2~3 名开发同时推进，环境搭建 + 品牌定制 / 预览 / UTM / 编辑器可同步进行。

5.2 Sprint 2：部署上线测试 (04-28 → 05-11)

Sprint Goal：Staging + Production 环境部署完成，全量功能测试通过，UAT 验收完成。

任务	估算	依赖
Staging 环境部署 (Docker + PostgreSQL + Redis + Nginx)	2d	Sprint 1 完成
Production 环境部署	2d	Staging 验证通过
DKIM / SPF / DMARC 域名验证配置	1d	域名 DNS 权限
端到端功能测试 (完整 Campaign 流程)	2d	Staging 部署
SES 发送压力测试 (1000+ 收件人批量发送)	1d	Staging 部署
邮件客户端兼容性测试 (Gmail / Outlook / Apple Mail / Yahoo)	2d	无
用户验收测试 (UAT， 邀请实际使用者操作验证)	2d	功能测试通过
Bug 修复与迭代	2d	测试反馈

5.3 Sprint 3：数据迁移 (05-12 → 05-25)

Sprint Goal：MailerLite 全量数据成功迁移到 Production，数据完整性校验通过。

任务	估算	依赖
----	----	----

从 MailerLite 导出全量数据（联系人、分组、退订列表）	1d	无
开发 CSV → Plunk 导入脚本（字段映射、分组关联、退订状态）	3d	数据导出
Staging 环境迁移验证（导入 + 数据完整性校验）	2d	导入脚本
Production 正式数据迁移（最新联系人 + 退订列表）	1d	Staging 验证通过
数据校验（总数、分组分布、退订列表比对）	1d	Production 迁移
历史邮件模板迁移（从 MailerLite 导出 HTML，导入 Plunk 模板库）	2d	模板库功能
SES Warm-up 逐步增加发送量	持续	Production 部署

5.4 Sprint 4 : 正式切换 (05-26 → 06-08)

Sprint Goal : DNS 切换完成，团队完成培训，MailerLite 正式下线，新系统进入稳定运行。

任务	估算	依赖
DNS 切换 (mail.infstones.com 指向新系统)	1d	数据迁移完成
Production 冒烟测试 (发送真实测试邮件验证全链路)	1d	DNS 切换
用户培训 & 使用文档编写	2d	无

首次真实 Campaign 发送（小批量验证）	1d	冒烟测试通过
MailerLite 正式下线（通知团队，停用旧系统）	1d	真实发送验证
监控告警配置（系统可用性、发送成功率、异常报警）	2d	Production 运行
上线首周 bug 修复与体验优化	持续	用户反馈

06-09 ~ 07-10：约 30 天缓冲期，充裕时间应对切换后的紧急问题和体验打磨。

6 数据迁移方案

6.1 迁移清单

数据类型	来源	说明
订阅者列表	MailerLite 导出 CSV	含 email、name、自定义字段、分组标签
退订列表	MailerLite 导出	合规必须，CAN-SPAM / GDPR 要求
分组/标签结构	MailerLite 导出	映射到 Plunk Segment
邮件模板 HTML	MailerLite 导出	作为 Plunk 模板库的初始素材

6.2 迁移步骤

a) Sprint 3 前半段 — Staging 验证：

- 从 MailerLite 导出全量 CSV 数据
- 开发 CSV → Plunk 导入脚本（处理字段映射、分组关联、退订状态）
- 在 Staging 环境验证数据完整性

b) Sprint 3 后半段 — Production 正式迁移：

- 重新导出 MailerLite 最新数据（确保增量更新）

- 执行 Production 导入
- 数据校验（总数、分组分布、退订列表比对）

7 风险与应对策略

风险	影响	概率	应对策略
Sprint 1 开发任务密集，功能开发不及时	高	中	多人并行开发，合理拆分任务；预览 / UTM / 编辑器可同步推进
Plunk 编辑器体验不达标	高	中	Sprint 1 中期做内部试用，如差距大则考虑集成 React Email 或 MJML 编辑器
AWS SES 发送配额不足	高	低	提前申请提高 SES 发送限额，确保满足 1W+ 收件人需求
SES Warm-up 周期影响	中	中	从 Sprint 2 开始逐步增加发送量，避免上线后突然大量发送触发限制
邮件客户端渲染不一致	中	中	Sprint 2 专项测试，使用 Litmus 或 Email on Acid 做跨客户端测试
数据迁移丢失或错误	高	低	Staging 先行验证，Production 迁移后做全量比对

8 成功指标

指标	目标	衡量方式
切换完成日期	≤ 2026-06-08（Sprint 4 结束）	Production 上线并完成首次真实发送
功能覆盖率	P0 100%	对照功能清单逐项验收
邮件送达率	≥ 95%	SES Dashboard 监控
用户满意度	UAT 通过，无阻塞性问题	UAT 验收报告
系统稳定性	上线首月无 P0 故障	监控告警 + 用户反馈

9 Feature Specifications

This section provides detailed specifications for all customization features, including user stories, acceptance criteria, technical design, and test cases. Engineering can develop and validate directly against this chapter.

9.1 Branding Customization

9.1.1 Problem Statement

Plunk ships with its own branding (logo, color scheme, product name, footer). Deploying it as-is would make users perceive it as a third-party tool rather than an InfStones internal system. All user-facing brand elements must be replaced with InfStones branding. The system remains fully English — no i18n or localization is required.

9.1.2 User Stories

#	User Story	Priority
US-1.1	As a marketing operator, I want the dashboard to display InfStones branding so that it feels like an internal tool.	P0
US-1.2	As a recipient, I want to see InfStones logo and brand colors in the	P0

	email footer so that the email looks official.	
--	--	--

9.1.3 Requirements & Acceptance Criteria

Req ID	Requirement	Acceptance Criteria
R-1.1	Replace Dashboard Branding	a) Top navbar logo replaced with InfStones Logo b) Browser tab title shows "InfStones Email Marketing" c) Favicon replaced with InfStones icon d) Login page displays InfStones branding
R-1.2	Color Scheme	a) Primary color replaced with InfStones brand color (from infstones.com) b) Buttons, links, highlights unified c) Existing UI layout and readability preserved
R-1.3	Email Footer Branding	a) Default footer shows InfStones company name and address b) Footer includes Unsubscribe link (CAN-SPAM compliant) c) Custom footer text supported
R-1.4	Domain Replacement	a) All internal URLs point to

		mail.infstones.co m b) API endpoint base URL configurable via environment variable
--	--	--

Acceptance Checklist:

- [] All pages show InfStones logo (not Plunk)
- [] Favicon replaced with InfStones icon
- [] Browser tab shows "InfStones Email Marketing"
- [] Email footer contains InfStones company name, address, and unsubscribe link
- [] Primary theme color unified to InfStones brand color
- [] Full English UI — no residual Plunk brand text

9.1.4 Technical Design

a) Brand Assets: Extract Logo SVG, brand color values, and font config from infstones.com.

b) Implementation: Use CSS Custom Properties to manage theme colors for easy future adjustments.

c) UI Language: Keep Plunk's native English UI. No i18n framework or translation files needed.

```

1 # Files to modify
2 apps/dashboard/public/           # Replace Logo SVG, Favicon
3 apps/dashboard/src/styles/       # Override CSS variables for brand
  colors
4 apps/dashboard/index.html        # Update <title> and meta tags
5 apps/api/src/templates/footer.ts # Update default email footer template
6 Global search-replace: "Plunk" → "InfStones Email Marketing"

```

9.1.5 Test Cases

TC ID	Scenario	Steps	Expected Result
TC-1.1	Logo replacement	Open Dashboard home page	InfStones logo displayed in navbar and login page
TC-1.2	Favicon & tab title	Open any Dashboard page in browser	Tab shows InfStones favicon and title

			"InfStones Email Marketing"
TC-1.3	Email footer	Send a campaign email, check footer	Footer shows InfStones name, address, and working unsubscribe link
TC-1.4	Brand color consistency	Navigate through all main pages	All buttons, links, and highlights use InfStones brand color
TC-1.5	No Plunk residue	Search entire UI for "Plunk" text	Zero occurrences of Plunk branding anywhere in the UI

9.2 Email Preview (Desktop + Mobile)

9.2.1 Problem Statement

MailerLite provides Desktop and Mobile preview modes, allowing operators to verify email rendering on different devices before sending. Plunk currently lacks built-in preview functionality — operators must send test emails and check their inbox to see the result, which is extremely inefficient.

9.2.2 User Stories

#	User Story	Priority
US-2.1	As a marketing operator, I want to preview the email in desktop view so that I can verify the layout looks correct on large screens.	P0

US-2.2	As a marketing operator, I want to preview the email in mobile view so that I can verify the layout renders correctly on phones.	P0
US-2.3	As a marketing operator, I want to toggle between desktop and mobile preview so that I can quickly compare both views.	P0
US-2.4	As a marketing operator, I want the preview to show personalized variables replaced with sample data so that I can see the real email experience.	P0

9.2.3 Requirements & Acceptance Criteria

Req ID	Requirement	Acceptance Criteria
R-2.1	Desktop Preview	a) Editor page provides a "Preview" button that opens the preview panel b) Desktop preview width: 600px (standard email width), centered c) Rendered in iframe for style isolation

R-2.2	Mobile Preview	a) Mobile preview width: 375px (iPhone SE standard) b) Renders responsive CSS media queries c) Wrapped in device frame for visual context
R-2.3	View Toggle	a) Desktop / Mobile toggle button (icon toggle) b) Smooth transition on switch (no page refresh) c) Default view: Desktop
R-2.4	Variable Substitution	a) Template variables (e.g. <code>{{name}}</code>) replaced with sample values b) Defaults: "John Doe" (name), "john@example.com" (email) c) Custom sample values supported
R-2.5	Real-time Preview	a) Preview syncs with editor content in real-time b) Update delay $\leq$ 500ms (debounce)

Acceptance Checklist:

- ☐ Editor page has a button to open/close preview panel
- ☐ Preview panel defaults to Desktop view (600px width)
- ☐ Clicking Mobile button switches to 375px width view

- [] Editing content updates preview within 500ms
- [] Preview correctly renders images, buttons, links, text styles
- [] Personalization variables show recognizable sample values
- [] Preview panel does not interfere with editor operations
- [] Empty content shows a friendly placeholder message

9.2.4 Interaction Flow

a) Entry Point: Campaign editor top toolbar — new "Preview" button (icon: eye).

b) Preview Panel: Right-side slide-out panel or full-screen modal. Side-by-side mode preferred (editor stays visible).

c) User Flow:

- User edits email content in editor
- Clicks "Preview" → preview panel opens, default Desktop view
- Clicks Mobile icon → switches to Mobile view
- Closes preview panel → returns to editor

9.2.5 Technical Design

a) Frontend Components:

```
1 | apps/dashboard/src/components/EmailPreview/
2 |   └─ EmailPreview.tsx           // Main preview container
3 |   └─ PreviewFrame.tsx          // iframe render component
4 |   └─ DeviceToggle.tsx          // Desktop/Mobile toggle buttons
5 |   └─ PreviewPlaceholder.tsx    // Empty state placeholder
```

b) Core Logic:

- Use `iframe.srcdoc` to inject editor HTML output
- Control viewport via CSS `width`; use `transform: scale()` to fit panel space
- Use `useDebounceValue` hook for 500ms debounce on editor content
- Variable substitution via regex matching `{{variable}}` with preset sample data

c) No backend changes required — preview is pure frontend rendering.

9.2.6 Test Cases

TC ID	Scenario	Steps	Expected Result
TC-2.1	Open preview panel	Click "Preview" button in editor	Preview panel opens, default

			Desktop view displayed
TC-2.2	Toggle device view	Click Mobile button	Preview area narrows to 375px, content reflows responsively
TC-2.3	Real-time update	Modify heading text in editor	Preview reflects change within 1 second
TC-2.4	Variable rendering	Insert {{name}} variable in editor	Preview shows "John Doe" as sample value
TC-2.5	Empty content	Open preview on a new blank Campaign	Placeholder message displayed (e.g. "Start editing to see preview")

9.3 Test Email Sending

9.3.1 Problem Statement

Operators need to send test emails to their own or colleagues' inboxes before mass sending, to verify the actual rendering in real email clients (Gmail, Outlook, Apple Mail). Browser preview alone cannot 100% reproduce client-specific rendering differences. Plunk does not provide this feature.

9.3.2 User Stories

#	User Story	Priority
US-3.1	As a marketing operator, I want to send a test email to my own address so that I can verify the email	P0

	looks correct in my actual inbox.	
US-3.2	As a marketing operator, I want to send test emails to multiple addresses so that colleagues can review before sending.	P0
US-3.3	As a marketing operator, I want the test email to include "[TEST]" in the subject so that I can distinguish test emails from real ones.	P1

9.3.3 Requirements & Acceptance Criteria

Req ID	Requirement	Acceptance Criteria
R-3.1	Send Test Email	a) Editor toolbar has "Send Test" button b) Click opens dialog for recipient input c) Supports multiple addresses (comma-separated), max 5 d) Delivered via SES within 5 seconds e) Success toast notification after send
R-3.2	Test Email Marker	a) Subject line auto-prefixed with "[TEST]" b) Test sends do not count toward Campaign analytics

		c) Test sends do not affect contact engagement data
R-3.3	Variable Substitution	a) Template variables replaced with sample values (same logic as preview) b) Or use real contact data if recipient exists in contact list
R-3.4	Recent Test Addresses	a) Remember recently used test addresses (localStorage) b) Auto-fill last used addresses on next open

Acceptance Checklist:

- ☐ Editor page has "Send Test" button
- ☐ Modal allows input of 1-5 email addresses
- ☐ Invalid email format shows validation error
- ☐ Test email received within 30 seconds of sending
- ☐ Test email subject has [TEST] prefix
- ☐ Test email content matches current editor content
- ☐ Personalization variables replaced with sample values
- ☐ Success toast shown after send
- ☐ Failure shows error message
- ☐ Reopening modal auto-fills last used addresses

9.3.4 Technical Design

a) New API Endpoint:

```

1 POST /api/v1/campaigns/:id/test-send
2
3 Request Body:
4 {
5   "recipients": ["test1@example.com", "test2@example.com"],
6   "subject_prefix": "[TEST] "
```

```

7 }
8
9 Response (200):
10 {
11   "success": true,
12   "sent_to": ["test1@example.com", "test2@example.com"],
13   "message_ids": ["ses-msg-id-1", "ses-msg-id-2"]
14 }
15
16 Response (400):
17 { "error": "INVALID_EMAIL", "details": "test1@invalid is not a valid
    email" }
18
19 Response (429):
20 { "error": "RATE_LIMITED", "message": "Max 10 test sends per hour per
    campaign" }

```

b) Backend Logic:

- Reuse Plunk's existing SES send module
- Render current Campaign HTML, substitute variables with sample values
- Rate limit: max 10 test sends per campaign per hour
- Test sends excluded from Campaign analytics

c) Frontend Components:

```

1 | apps/dashboard/src/components/TestSend/
2 | └─ TestSendButton.tsx    // Trigger button
3 | └─ TestSendModal.tsx    // Modal (email input + send)
4 | └─ TestSendHistory.tsx  // Optional: recent test send log

```

9.3.5 Test Cases

TC ID	Scenario	Steps	Expected Result
TC-3.1	Send single test email	Enter one valid email, click Send	Email received within 30s, subject has [TEST] prefix
TC-3.2	Send to multiple addresses	Enter 3 valid emails, click Send	All 3 inboxes receive the test email
TC-3.3	Invalid email validation	Enter "abc@" as email	Validation error shown, send blocked
TC-3.4	Exceed max recipients	Attempt to enter 6th email	UI prevents adding more

		address	than 5 addresses
TC-3.5	Rate limiting	Send >10 test emails in 1 hour for same campaign	429 error returned, rate limit message shown
TC-3.6	Address memory	Send test, close modal, reopen modal	Last used email addresses auto-filled

9.4 UTM Parameter Management

9.4.1 Problem Statement

Marketing emails need UTM parameters appended to all links for Google Analytics traffic tracking. Currently operators must manually add UTM to each link in MailerLite, which is error-prone and inefficient. A centralized UTM management panel that auto-appends parameters to all links is required.

9.4.2 User Stories

#	User Story	Priority
US-4.1	As a marketing operator, I want to configure UTM parameters for each campaign so that I can track traffic sources in Google Analytics.	P0
US-4.2	As a marketing operator, I want UTM parameters to be automatically appended to all links in the email so that I do not need to manually edit each link.	P0

US-4.3	As a marketing operator, I want to set default UTM values that apply to all campaigns so that I do not need to configure from scratch each time.	P1
--------	--	----

9.4.3 Requirements & Acceptance Criteria

Req ID	Requirement	Acceptance Criteria
R-4.1	UTM Config UI	a) Campaign details page has "UTM Settings" section b) Fields: - utm_source , - utm_medium , - utm_campaign , - utm_term (optional), - utm_content (optional) c) utm_source default: "email" d) utm_medium default: "newsletter" e) utm_campaign auto-filled with Campaign name (editable)
R-4.2	Auto-Append Logic	a) On send, all <code><a href></code> links get UTM params appended b) Skip: <code>mailto:</code> , <code>#</code> anchors, unsubscribe links c) Preserve existing

		params (do not overwrite manually set values) d) Handle existing query strings correctly (& vs ?)
R-4.3	Global Defaults	a) Settings page has "Default UTM Parameters" config b) New campaigns inherit global defaults c) Campaign-level config overrides global defaults
R-4.4	UTM Persistence	a) UTM config saved in Campaign record (database) b) Editing existing Campaign correctly restores UTM values

Acceptance Checklist:

- ☐ Campaign details page shows UTM parameter config section
- ☐ `utm_medium` defaults to "email"
- ☐ `utm_campaign` auto-fills with Campaign name
- ☐ Sent emails have UTM params on all external links
- ☐ `mailto:` and unsubscribe links excluded from UTM
- ☐ URLs with existing query strings use `&` correctly
- ☐ Empty UTM fields are not appended

9.4.4 Technical Design

a) Database Schema Change:

```

1  -- Add UTM fields to campaigns table (JSONB type)
2  ALTER TABLE campaigns ADD COLUMN utm_params JSONB DEFAULT NULL;
3
4  -- Example data:

```

```
5 -- {"utm_source": "newsletter", "utm_medium": "email", "utm_campaign":
  "april-update", "utm_term": "", "utm_content": ""}
```

b) API Change:

```
1 PUT /api/v1/campaigns/:id
2 {
3   ...existing fields,
4   "utm_params": {
5     "utm_source": "newsletter",
6     "utm_medium": "email",
7     "utm_campaign": "april-update",
8     "utm_term": "",
9     "utm_content": ""
10  }
11 }
```

c) UTM Append Logic (applied at send time):

```
1 // packages/api/src/utls/utm.ts
2 function appendUtmParams(html: string, utmParams: UtmParams): string {
3   // 1. Parse HTML, find all <a href="..."> tags
4   // 2. Filter: skip mailto:, #anchors, unsubscribe links
5   // 3. For each qualifying link, append UTM query params
6   // 4. Return modified HTML
7 }
```

d) Frontend Components:

```
1 apps/dashboard/src/components/UTM/
2 | UtmConfigPanel.tsx // UTM parameter input form
3 | UtmPreviewToggle.tsx // Toggle UTM display in preview
4 | UtmLinkTooltip.tsx // Hover tooltip showing full UTM URL
```

9.4.5 Test Cases

TC ID	Scenario	Steps	Expected Result
TC-4.1	UTM auto-append	Configure UTM params, send email with 3 external links	All 3 links include UTM parameters
TC-4.2	Exclude special links	Email contains mailto: and unsubscribe links	These links have no UTM params
TC-4.3	Existing query string	Link is https://example.com?ref=abc	UTM appended with &: ?ref=abc&utm_source=...

TC-4.4	Empty UTM fields	Only fill source and medium, leave others blank	Only non-empty fields appended
TC-4.5	utm_campaign auto-fill	Create campaign named "April Update"	utm_campaign auto-fills as "april-update"
TC-4.6	UTM persistence	Save UTM config, close campaign, reopen it	UTM values correctly restored

9.5 UTM Email Preview

9.5.1 Problem Statement

After configuring UTM parameters, operators cannot verify whether links are correctly appended before sending. They need to see the full URL with UTM parameters in the preview to avoid incorrect tracking data in Google Analytics.

9.5.2 User Stories

#	User Story	Priority
US-5.1	As a marketing operator, I want to see the full URL with UTM parameters when hovering over a link in preview so that I can verify UTM configuration.	P0
US-5.2	As a marketing operator, I want a dedicated "UTM Preview" panel that lists all links with their complete UTM URLs so	P0

	that I can quickly audit all links at once.	
--	---	--

9.5.3 Requirements & Acceptance Criteria

Req ID	Requirement	Acceptance Criteria
R-5.1	Hover Preview	a) Hovering a link in email preview shows a tooltip b) Tooltip displays the full URL with UTM params c) Long URLs wrap automatically
R-5.2	UTM Links Summary Panel	a) Preview panel has a "Links" tab at the bottom b) Lists all external links with complete UTM URLs c) One-click copy for individual links d) Updates in real-time when UTM config changes
R-5.3	Real-time Sync	a) Changing UTM config updates preview links within 500ms b) No manual refresh needed

Acceptance Checklist:

- ☐ Hovering links in preview shows full UTM URL tooltip
- ☐ "Links" tab lists all external links with UTM params
- ☐ One-click copy works for each link

- [] Changing UTM config updates preview links within 500ms
- [] Empty UTM params result in no extra parameters on links

9.5.4 Technical Design

a) Dependencies: Requires Feature 9.2 (Email Preview) and Feature 9.4 (UTM Management) to be completed first.

b) Implementation: Apply UTM append logic (reuse R-4.2 function) during preview rendering phase. Extract all links to generate the summary list.

9.5.5 Test Cases

TC ID	Scenario	Steps	Expected Result
TC-5.1	UTM tooltip	Configure UTM, hover a link in preview	Tooltip shows full URL with UTM params
TC-5.2	Links summary panel	Open Links tab in preview panel	All external links listed with complete UTM URLs
TC-5.3	Copy link	Click copy icon next to a link	Full UTM URL copied to clipboard
TC-5.4	Real-time update	Change utm_source value	Preview links and Links tab update within 1 second
TC-5.5	No UTM configured	Leave all UTM fields empty, check preview	Links shown without any UTM parameters

9.6 Editor Enhancement

9.6.1 Problem Statement

Plunk's built-in drag-and-drop editor supports basic block types (text, image) but lacks Social Links, rich Button styles, and Divider components that MailerLite users rely on. The drag-and-

drop interaction also needs improvement for a smoother editing experience.

9.6.2 User Stories

#	User Story	Priority
US-6.1	As a marketing operator, I want to add social media icon links (Twitter/X, LinkedIn, Telegram, Discord, etc.) so that emails include our social presence.	P0
US-6.2	As a marketing operator, I want to customize button styles (color, size, border radius, full-width) so that CTAs stand out appropriately.	P0
US-6.3	As a marketing operator, I want to add visual dividers between sections so that the email layout is clean and well-structured.	P0
US-6.4	As a marketing operator, I want smooth drag-and-drop when reordering blocks so that editing is intuitive and fast.	P0

9.6.3 Requirements & Acceptance Criteria

Req ID	Requirement	Acceptance Criteria
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R-6.1	Social Links Block	a) New "Social Links" block type in sidebar b) Preset icons: Twitter/X, LinkedIn, Telegram, Discord, GitHub, YouTube, Instagram, Facebook c) Each icon URL configurable d) Horizontal layout (center/left/right alignment) e) Icon size: small (24px), medium (32px), large (48px) f) Output HTML uses img tags (not SVG) for email client compatibility
R-6.2	Button Style Enhancement	a) Button block gets a style config panel b) Configurable: bg color, text color, border radius (0/4/8/20px), width (auto/full-width) c) Font size (14/16/18px), font weight (normal/bold) d) 3 presets: Primary (filled), Secondary (outlined), Text-only e) Hover state color config (for supporting email clients)
R-6.3	Divider Block	a) New "Divider" block type

		b) Configurable: line style (solid/dashed/dotted), color, thickness (1-3px) c) Configurable: top/bottom padding d) Default: 1px solid #e0e0e0, 16px padding
R-6.4	Drag-and-Drop Optimization	a) Blue insertion indicator line during drag b) Dragged element shows semi-transparent preview c) Smooth animation on drop (200ms transition) d) No visible stuttering or flickering

Acceptance Checklist:

- ☐ Block panel shows Social Links and Divider as new block types
- ☐ Social Links supports at least 8 mainstream platform icons
- ☐ Users can select/deselect platforms and configure link URLs
- ☐ Button supports at least 3 styles and 4 border-radius options
- ☐ Divider supports 3 line styles
- ☐ Drag indicator shows insertion position
- ☐ Drag-and-drop reorder is smooth with no lag
- ☐ All new blocks render correctly in preview and actual sent emails
- ☐ All new blocks render correctly in Gmail, Outlook, Apple Mail

9.6.4 Technical Design

a) Block Registration:

- Plunk editor uses a block registration pattern. New blocks must be registered in the Block Registry.

- Each block includes: Editor Component (edit mode), Render Function (HTML output), Settings Panel (config UI).

```

1 apps/dashboard/src/components/Editor/blocks/
2   └─ SocialLinksBlock/
3       └─ SocialLinksEditor.tsx    // Edit mode component
4       └─ SocialLinksSettings.tsx // Config panel
5       └─ SocialLinksRenderer.ts  // HTML output
6       └─ icons/                  // Platform SVG icons
7   └─ DividerBlock/
8       └─ DividerEditor.tsx
9       └─ DividerSettings.tsx
10      └─ DividerRenderer.ts
11  └─ ButtonBlock/                // Enhance existing Button
12      └─ ButtonStylePresets.tsx  // Style preset selector
13      └─ ButtonSettings.tsx      // Enhanced config panel

```

b) Email Client Compatibility:

- Social icons: use hosted image URLs (some clients don't support inline SVG)
- Buttons: use `<table>` + `<td>` bulletproof button technique (Outlook compatibility)
- Dividers: use `<table>` + `border` (not `<hr>` — better cross-client rendering)

9.6.5 Test Cases

TC ID	Scenario	Steps	Expected Result
TC-6.1	Add Social Links	Drag Social Links block into editor	Default 4 platform icons displayed (Twitter, LinkedIn, Discord, Telegram)
TC-6.2	Configure Social Links	Open config panel, remove Discord, add GitHub	Editor and preview update accordingly
TC-6.3	Button style switch	Select Outlined + large radius + brand color	Button updates to outlined + rounded style in real-time
TC-6.4	Insert Divider	Insert Divider between two	Gray solid line divider displayed

		text blocks	
TC-6.5	Drag reorder	Drag 3rd block to 1st position	Block order correctly adjusted with animation
TC-6.6	Outlook compatibility	Send email with Social Links + Button to Outlook	Icons and buttons render correctly

9.7 Email Template Library

9.7.1 Problem Statement

Creating campaigns from scratch every time is inefficient. MailerLite provides a rich template library for one-click start. Plunk has no template library feature — one needs to be built.

9.7.2 User Stories

#	User Story	Priority
US-7.1	As a marketing operator, I want to browse a template gallery when creating a campaign so that I can start with a professional layout instead of a blank page.	P0
US-7.2	As a marketing operator, I want to preview a template before applying it so that I can choose the best fit.	P0
US-7.3	As a marketing operator, I want to save	P0

	my own campaigns as reusable templates so that I can reuse successful designs.	
--	--	--

9.7.3 Requirements & Acceptance Criteria

Req ID	Requirement	Acceptance Criteria
R-7.1	Template Gallery Page	a) Dashboard nav adds "Templates" menu item b) Templates displayed as cards (thumbnail + name + category tag) c) Filter by category: Newsletter, Product Update, Promotion, Event, Announcement d) Search by template name
R-7.2	Preset Templates (10+)	a) Newsletter x3 (minimal / image-text / magazine) b) Product Update x2 (feature launch / version update) c) Promotion x2 (flash sale / seasonal promo) d) Event x2 (online invitation / meetup) e) Announcement x1 (general) f) All use InfStones brand colors and logo g) All responsive for Desktop and Mobile

R-7.3	Template Application	a) Campaign creation offers "Start from template" or "Start blank" b) Selected template loads into editor c) Content fully editable after loading d) If editor has existing content, confirm before overwriting
R-7.4	Custom Templates	a) Editor has "Save as Template" button b) Save requires name and category selection c) Custom templates shown in gallery with "Custom" tag d) Custom templates can be deleted (presets cannot)

Acceptance Checklist:

- ☐ Campaign creation flow starts with template selection
- ☐ Gallery shows at least 10 preset templates
- ☐ Template cards display thumbnail, name, category
- ☐ Selecting template loads content into editor
- ☐ Category filter works
- ☐ "Save as Template" button available in editor
- ☐ Custom templates appear in gallery
- ☐ Custom templates can be deleted and renamed
- ☐ "Blank" option available for starting from scratch

9.7.4 Technical Design

a) Database Schema:

```

1 CREATE TABLE email_templates (
2   id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),
3   name        VARCHAR(255) NOT NULL,
4   category    VARCHAR(50) NOT NULL, -- 'newsletter',
   'product_update', 'promotion', 'event', 'custom'
5   content     JSONB NOT NULL,      -- Editor block data (JSON
format)
6   html        TEXT,                -- Pre-rendered HTML (for
thumbnail and quick preview)
7   thumbnail   TEXT,                -- Thumbnail URL or base64
8   is_preset   BOOLEAN DEFAULT FALSE, -- true = system preset, false =
user custom
9   project_id  UUID REFERENCES projects(id),
10  created_at  TIMESTAMP DEFAULT NOW(),
11  updated_at  TIMESTAMP DEFAULT NOW()
12 );
13
14 CREATE INDEX idx_templates_category ON email_templates(category);
15 CREATE INDEX idx_templates_project ON email_templates(project_id);

```

b) API Endpoints:

```

1 GET    /api/v1/templates          # List (supports ?category=
filter)
2 GET    /api/v1/templates/:id      # Get single template
3 POST   /api/v1/templates          # Create custom template
4 PUT    /api/v1/templates/:id      # Update custom template
5 DELETE /api/v1/templates/:id      # Delete custom template
(custom only)
6 POST   /api/v1/templates/:id/duplicate # Duplicate template

```

c) Frontend Components:

```

1 apps/dashboard/src/components/Templates/
2 └─ TemplateGallery.tsx      // Gallery main page
3 └─ TemplateCard.tsx        // Template card component
4 └─ TemplateCategoryFilter.tsx // Category filter
5 └─ SaveAsTemplateModal.tsx  // Save-as-template dialog
6 └─ TemplatePreview.tsx     // Template detail preview

```

9.7.5 Test Cases

TC ID	Scenario	Steps	Expected Result
TC-7.1	Create from template	Select "Classic Newsletter" template	Editor loads template content, fully editable
TC-7.2	Category filter	Click "Promotion" category	Only promotion templates shown
TC-7.3	Save custom template	Edit email, click "Save as	Template appears in

		Template"	gallery under Custom category
TC-7.4	Delete custom template	Delete a custom template	Template removed from gallery; preset templates show no delete option
TC-7.5	Blank template	Select "Blank" option	Empty editor opens

9.8 Dashboard Performance Overview

9.8.1 Problem Statement

Plunk's current Dashboard provides basic send statistics but lacks KPI summary cards, trend charts, and cross-campaign comparison. The marketing team needs a higher-density Dashboard to quickly assess overall email marketing performance and trends.

9.8.2 User Stories

#	User Story	Priority
US-8.1	As a marketing operator, I want to see key KPIs at a glance (total subscribers, open rate, click rate, bounce rate) so that I can quickly assess overall performance.	P0
US-8.2	As a marketing operator, I want to see trend charts for open rate and click rate over time so that I can identify performance patterns.	P0

US-8.3	As a marketing operator, I want to compare metrics across campaigns so that I can identify which campaigns performed best.	P0
--------	--	----

9.8.3 Requirements & Acceptance Criteria

Req ID	Requirement	Acceptance Criteria
R-8.1	KPI Cards	a) Dashboard top shows 4 KPI cards: Total Subscribers, Avg Open Rate, Avg Click Rate, Bounce Rate b) Each card shows current value + month-over-month change (green↑/red↓) c) Data based on last 30 days of campaigns
R-8.2	Trend Chart	a) Line chart showing last 12 campaigns' Open Rate and Click Rate b) X-axis: campaign send date; Y-axis: percentage c) Hover tooltip shows exact values d) Time range toggle: Last 30 Days / Last 90 Days / Last 6 Months

R-8.3	Campaign Comparison Table	a) Table of last 10 campaigns with: sent count, open rate, click rate, unsubscribes b) Sortable by column c) Highest/lowest values highlighted d) Click campaign name to navigate to details
-------	---------------------------	---

Acceptance Checklist:

- ☐ Dashboard top shows 4 KPI cards
- ☐ KPI cards show current month data + month-over-month comparison
- ☐ Trend chart correctly renders last 30 days of data
- ☐ Trend chart supports 7/30/90 day toggle
- ☐ Campaign comparison table lists last 10 campaigns
- ☐ Table supports sorting by open rate / click rate
- ☐ Empty state guidance shown when no campaign data exists

9.8.4 Technical Design

a) API Endpoint:

```

1 GET /api/v1/analytics/overview?period=30d
2
3 Response:
4 {
5   "kpi": {
6     "total_sent": 5420,
7     "total_sent_prev": 4800,
8     "avg_open_rate": 0.284,
9     "avg_open_rate_prev": 0.261,
10    "avg_click_rate": 0.052,
11    "avg_click_rate_prev": 0.048,
12    "unsubscribe_rate": 0.003,
13    "unsubscribe_rate_prev": 0.004
14  },
15  "trend": [
16    {"date": "2026-03-15", "sent": 0, "opens": 0, "clicks": 0},
17    {"date": "2026-03-16", "sent": 1200, "opens": 340, "clicks": 62}
18  ],
19  "campaigns": [
20    {
21      "id": "...",
22      "name": "March Newsletter",
23      "sent_at": "2026-03-20T10:00:00Z",
24      "total_sent": 1200,

```

```

25     "open_rate": 0.31,
26     "click_rate": 0.06,
27     "unsubscribes": 2
28   }
29 ]
30 }

```

b) Frontend: Use [Recharts](#) (React charting library) for trend charts.

```

1 apps/dashboard/src/components/Dashboard/
2 └─ KpiCards.tsx           // 4 KPI cards
3 └─ TrendChart.tsx        // Trend line chart
4 └─ CampaignComparison.tsx // Campaign comparison table
5 └─ PeriodSelector.tsx     // Time range toggle

```

c) Performance: Cache aggregated results in Redis (TTL 5 minutes) to avoid heavy queries on every Dashboard load.

9.8.5 Test Cases

TC ID	Scenario	Steps	Expected Result
TC-8.1	KPI cards display	Open Dashboard home page	4 KPI cards shown with current values and month-over-month arrows
TC-8.2	Trend chart	View trend chart on Dashboard	Line chart shows Open Rate and Click Rate over last 30 days
TC-8.3	Time range toggle	Switch to 90 days view	Chart data range expands to 90 days
TC-8.4	Campaign comparison sort	Click "Open Rate" column header	Table sorts by open rate descending
TC-8.5	Empty state	Dashboard with no campaigns sent yet	Friendly empty state with guidance message

9.9 Data Migration Tooling

9.9.1 Problem Statement

Switching from MailerLite to the new system requires complete migration of all subscriber data, group structures, and unsubscribe lists. Data loss or errors during migration directly impacts compliance (CAN-SPAM / GDPR require preserving unsubscribe records) and business continuity.

9.9.2 User Stories

#	User Story	Priority
US-9.1	As a system administrator, I want a migration script that imports MailerLite CSV data into the new system so that I can complete data migration.	P0
US-9.2	As a system administrator, I want to run a data validation script after migration so that I can confirm data integrity.	P0
US-9.3	As a system administrator, I want the unsubscribe list to be fully migrated so that we remain compliant.	P0

9.9.3 Requirements & Acceptance Criteria

a) Migration Script:

- Read MailerLite exported CSV files
- Field mapping: email → email, name → metadata.name, custom fields → metadata

- Group mapping: MailerLite Group → Plunk Segment/Tag
- Unsubscribe status: MailerLite unsubscribed contacts → Plunk unsubscribe list
- Deduplication: same email keeps latest record only
- Dry-run mode: output mapping report without writing to database

b) Data Validation Script:

- Compare MailerLite export count vs Plunk import count
- Compare per-group counts
- Compare unsubscribe list count
- Output difference report

Acceptance Checklist:

- [] Migration script supports Dry-run mode
- [] All MailerLite contacts successfully imported into Plunk
- [] Group/tag structure correctly mapped
- [] Unsubscribe list fully migrated (count matches)
- [] Custom fields (name, etc.) correctly preserved
- [] Dedup logic correct (no duplicate emails)
- [] Validation script outputs PASS/FAIL report
- [] Staging environment validated before Production migration

9.9.4 Technical Design

```

1 scripts/migration/
2 |— migrate.ts          // Main migration script
3 |— validate.ts        // Data validation script
4 |— field-mapping.ts   // Field mapping config
5 |— csv-parser.ts      // CSV parser (handles MailerLite export
format)
6 |— README.md          // Usage instructions
7
8 # Usage:
9 npx ts-node scripts/migration/migrate.ts --file=contacts.csv --dry-run
10 npx ts-node scripts/migration/migrate.ts --file=contacts.csv --
env=staging
11 npx ts-node scripts/migration/validate.ts --source=contacts.csv --
env=staging

```

9.9.5 Test Cases

TC ID	Scenario	Steps	Expected Result
TC-9.1	Dry-run mode	Run migrate.ts --dry-run	Mapping report output, no data

			written to DB
TC-9.2	Normal import	Import 1000 MailerLite contacts	1000 new contacts in Plunk
TC-9.3	Unsubscribe migration	Import CSV with 50 unsubscribed records	50 contacts marked as unsubscribed in Plunk
TC-9.4	Deduplication	CSV contains 3 duplicate emails	Only 1 record imported per email (latest)
TC-9.5	Data validation	Run validate.ts after import	Comparison report shows PASS
TC-9.6	Group mapping	CSV contains 3 MailerLite Groups	3 corresponding Segments created in Plunk

9.10 User Management

9.10.1 Problem Statement

The current system has no user management capability. There is no login/registration, no user list, and no way to control who can access the system. As the team grows, we need basic user management to control system access and distinguish admin operations (e.g., managing users) from regular usage.

9.10.2 User Stories

#	User Story	Priority
US-10.1	As an admin, I want to add users by email so that new team members can access the system.	P0

US-10.2	As an admin, I want to remove users so that former team members lose access immediately.	P0
US-10.3	As an admin, I want to assign Admin or Member roles so that only admins can manage users.	P0
US-10.4	As a user, I want to see my identity (avatar/email) in the top-right corner so that I know which account I am logged in with.	P0
US-10.5	As a member, I want to use all system features except user management so that I can do my daily work without restrictions.	P0

9.10.3 Requirements & Acceptance Criteria

Req ID	Requirement	Acceptance Criteria
R-10.1	Users Page in Settings	a) Settings navigation adds a "Users" tab b) Users page displays a table with columns: Email, Role, Actions c) Only users with Admin role can see and access the Users tab d) Members who

		navigate to the Users URL directly are redirected or shown an access denied message
R-10.2	Add User	a) "Add User" button opens a form/dialog b) Required field: Email address (validated format) c) Role selector: Admin or Member (default: Member) d) After adding, user appears in the table immediately e) Added user can log in to the system with their email (no invitation flow needed)
R-10.3	Remove User	a) Each row has a "Remove" action button b) Confirmation dialog before removal c) Removed user can no longer log in d) Cannot remove yourself (prevent admin logout)
R-10.4	Role Assignment	a) Two roles: Admin (full access) and Member (no Users page access) b) Admin can change any user's role via the Users table

		c) At least one Admin must exist at all times (prevent last-admin demotion) d) Role change takes effect on the user's next page load
R-10.5	Current User Indicator	a) Top-right corner of the app header shows the current logged-in user b) Display: user email (or first letter avatar + email) c) Clicking shows a dropdown with "Log out" option d) Visible on all pages
R-10.6	Authentication (Google SSO)	a) Login page displays "Sign in with Google" button (Google brand style), no email/password inputs b) Only <code>@infstones.com</code> Google Workspace accounts are allowed; other domains show error: "Only <code>@infstones.com</code> accounts are allowed" c) Google login email is matched against Users list — unmatched accounts are rejected: "No account found"

		d) On successful login, retrieve Google profile (avatar, display name) and use as user indicator e) Session persists across browser refreshes f) Logout clears session and redirects to login page Demo Implementation: g) Login page shows "Sign in with Google" button h) Clicking opens simulated Google account picker with 2–3 preset <code>@infstones.com</code> accounts (with avatars) i) Selecting a registered account logs in; selecting unregistered or non-infstones account shows appropriate error
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9.10.4 Technical Design

a) Data Model:

Field	Type	Description
id	UUID	Primary key
email	String (unique)	User login identifier

google_id	String (unique)	Google account identifier
avatar_url	String (nullable)	Google profile avatar URL
display_name	String (nullable)	Google profile display name
role	Enum: admin member	Determines access level
createdAt	DateTime	Account creation timestamp

b) Permission Matrix:

Feature	Admin	Member
Dashboard	✓	✓
Campaigns	✓	✓
Subscribers	✓	✓
Groups	✓	✓
Templates	✓	✓
Settings (General/Sending/Import)	✓	✓
Settings > Users	✓	⌵

c) API Endpoints:

Method	Endpoint	Description	Auth
POST	/api/auth/google	Google OAuth callback — verify token, match email against	Public

		Users list, check @infstones.com domain	
POST	/api/auth/logout	Clear session	Authenticated
GET	/api/users	List all users	Admin only
POST	/api/users	Add new user	Admin only
PATCH	/api/users/:id	Update user role	Admin only
DELETE	/api/users/:id	Remove user	Admin only
GET	/api/auth/me	Get current user info	Authenticated

d) UI Components:

- Settings > Users tab: table with email, role badge (Admin=indigo, Member=gray), and action buttons
- Add User dialog: email input + role selector + "Add" button
- Header user indicator: Google avatar (or first-letter fallback) + email text + dropdown menu (Log out)
- Login page: centered card with "Sign in with Google" button (Google brand colors and logo). Demo: clicking opens simulated Google account picker overlay with preset @infstones.com accounts
- Header user indicator: Google avatar (real photo from Google profile) + display name/email + dropdown menu (Log out)

9.10.5 Test Cases

TC ID	Scenario	Input	Expected Result
TC-10.1	Add user	Admin enters valid email, selects Member role	User appears in table, can log in
TC-10.2	Add duplicate email	Admin enters email that already exists	Error message: "User already exists"

TC-10.3	Remove user	Admin clicks Remove on a Member	Confirmation dialog → user removed from table → user cannot log in
TC-10.4	Prevent self-removal	Admin tries to remove themselves	Remove button disabled or error: "Cannot remove yourself"
TC-10.5	Change role	Admin changes a Member to Admin	Role badge updates, user gains Users tab access on next load
TC-10.6	Last admin protection	Only admin tries to demote themselves to Member	Error: "At least one admin required"
TC-10.7	Member access restriction	Member navigates to /settings/users	Redirected to Dashboard or shown "Access Denied"
TC-10.8	Current user indicator	Any logged-in user on any page	Top-right shows avatar + email, dropdown has "Log out"
TC-10.9	Google SSO Login	Click "Sign in with Google", select @infstones.com account that exists in Users list	Redirected to Dashboard, Google avatar shown in header, session persists on refresh

TC-10.11	Non-infstones domain rejected	Click "Sign in with Google", select non-@infstones.com account	Error: "Only @infstones.com accounts are allowed"
TC-10.12	Unregistered infstones account rejected	Click "Sign in with Google", select @infstones.com account NOT in Users list	Error: "No account found with this email address"
TC-10.13	Demo account picker	Click "Sign in with Google" on demo	Simulated Google account picker appears with 2–3 preset accounts with avatars
TC-10.10	Logout	Click "Log out" from dropdown	Session cleared, redirected to login page

10 Open Questions

#	Question	Owner	Blocking Level	Status
OQ-1	Where to obtain InfStones brand specification files (Logo, color values, fonts)?	Design	Before Sprint 1 start	Open

OQ-2	Deployment target environment: AWS EC2 / ECS / EKS? Need DevOps confirmation.	DevOps	Before Sprint 2 start	Open
OQ-3	Current AWS SES sending quota? Need to request increase?	DevOps	Sprint 1	Open
OQ-4	Who manages DNS for mail.infstones.com? Who to coordinate for the switch?	DevOps	Before Sprint 4 start	Open
OQ-5	What custom fields exist in MailerLite currently? Need full field list for migration mapping.	Marketing	Before Sprint 3 start	Open
OQ-6	Which drag library does Plunk's editor use? (react-dnd / @dnd-kit / custom)	Engineering	Sprint 1	Open

	Need to determine enhancement approach.			
OQ-7	Should Social Links icons be hosted on InfStones CDN or inlined as SVG?	Engineering	Non-blocking	Open
OQ-8	Who are the UAT testers? Need to confirm participant list in advance.	PM	Before Sprint 2 start	Open

11 Non-Goals

Item	Reason
A/B Testing	Current send volume (monthly <10 campaigns) insufficient for statistical significance. Plan for P2 phase.
Auto Resend to Non-Openers	P2 phase feature, not required for core replacement.
Advanced Automation Sequences (Drip Campaigns)	Current usage is manual trigger only. Automation is P2.
Landing Page Builder	Not within current MailerLite usage scope.

CRM Integration	Beyond email marketing system scope.
Multi-Tenancy	Internal use only at InfStones, multi-tenancy not needed.